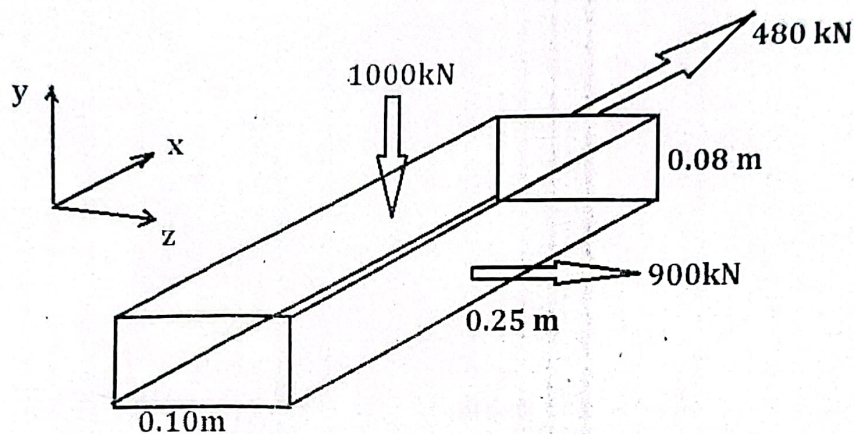


INSTRUCTIONS

(i) Attempt all the questions

(ii) All parameters have their usual meaning unless otherwise specified.

1. A rectangular steel block is subjected to three mutually perpendicular forces as shown in Fig. Q1. Assuming $E = 200\text{GPa}$ and $\nu = 0.25$,
 - (a) Determine the strains in the direction of each force (3 marks)
 - (b) Calculate the change in volume of the block (2 marks)
 - (c) Determine the bulk modulus (K) and modulus of rigidity (G). (2 marks)



2. A tension specimen of circular cross section tapers uniformly from 20mm to 16mm diameter over a gauge length of 200mm as shown in Fig. Q.1b. When an axial force of 40kN is applied, the extension measured on this gauge length is 0.4mm.
 - (i) Find the modulus of elasticity of the material. [3 marks]
 - (ii) Compare the strain energy in this specimen with that in a specimen of the same material of uniform diameter 18mm carrying the same load. [2 marks]

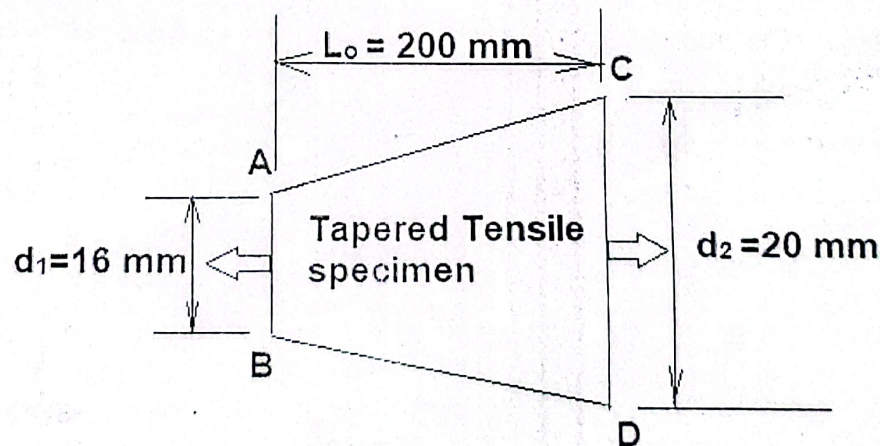


Fig. Q.2.

3. Find the forces in all the members of the truss shown in Fig. Q. 3 using the method of joints.
Qn from Engineering Mechanics by S.S. Bhavikatti
 Answers remain the same. (10 Marks)

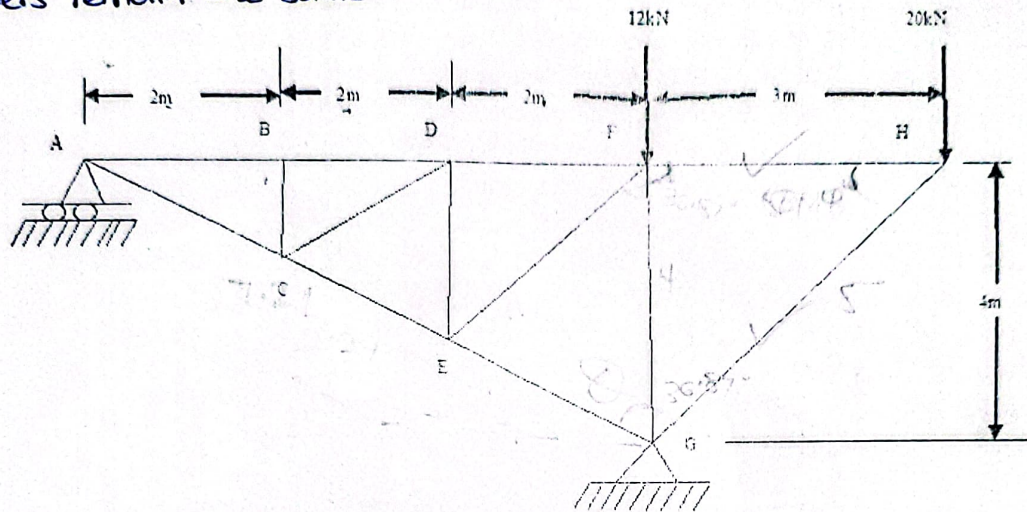


Fig. Q. 3

4. A compressed-air tank having an inner diameter of 5.5 m and a wall thickness of 45 mm is formed by welding two steel hemispheres as shown in Fig. Q. 4
- If the allowable tensile stress in the steel is 93 MPa, what is the maximum permissible air pressure P_a in the tank? (1 mark)
 - If the allowable shear stress in the steel is 42 MPa, what is the maximum permissible pressure P_b ? (1 mark)
 - If the normal strain at the outer surface of the tank is not to exceed 0.0003, what is the maximum permissible pressure P_c ? (Assume that Hooke's law is valid and that the modulus of elasticity for the steel is 210 GPa and Poisson's ratio is 0.28.) (3 marks)
 - Tests on the welded seam show that failure occurs when the tensile load on the welds exceeds 7.5 MN per meter of weld. If the required factor of safety against failure of the weld is 2.5, what is the maximum permissible pressure P_d ? (2 marks)
 - Considering the four preceding factors, what is the allowable pressure in the tank? (1 mark)

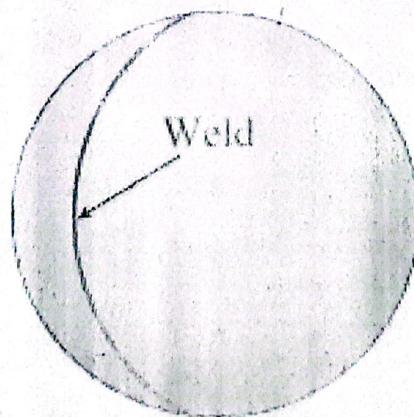


Fig. Q. 4

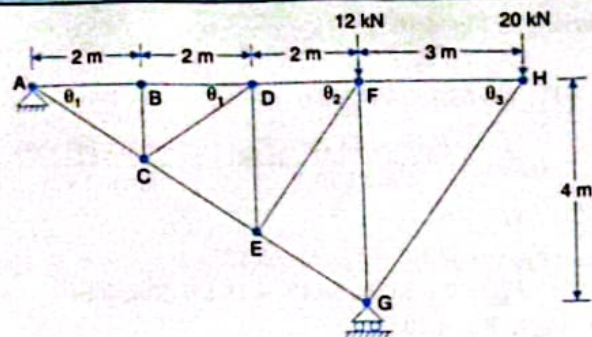


Fig. 4.11 (a)

Joint by joint analysis is carried out as given below and the joint forces are marked in Fig. 4.11 (b). Then the nature of the force in each member is determined.

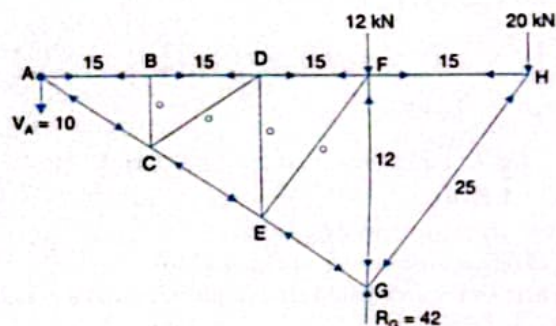


Fig. 4.11 (b)

Joint H: $\Sigma F_V = 0 \rightarrow$
 $F_{HG} \sin \theta_3 = 20$

$\therefore F_{HG} = \frac{20}{\sin 53.13^\circ} = 25 \text{ kN (Compression)}$

$\Sigma F_H = 0 \rightarrow$
 $F_{HF} - F_{HG} \cos \theta_3 = 0$
 $\therefore F_{HF} = 25 \cos 53.13^\circ = 15 \text{ kN (Tension)}$

Now, reactions at supports are to be found.

$\Sigma M_A = 0 \rightarrow$
 $-R_G \times 6 + 20 \times 9 + 12 \times 6 = 0$

$\therefore R_G = 42 \text{ kN}$

$\Sigma F_V = 0 \rightarrow$

$V_A + R_G - 12 - 20 = 0$

$V_A = 32 - 42, \text{ since } R_G = 42 \text{ kN}$
 $= -10 \text{ kN}$
 $= 10 \text{ kN } \downarrow$

$\Sigma F_H = 0 \rightarrow H_A = 0.$

Hence reactions are as shown in Fig. 4.11 (b)

Joint A: $\Sigma F_V = 0 \rightarrow$

$F_{AC} \sin \theta_1 - 10 = 0$

$\therefore F_{AC} = \frac{10}{\sin 33.69^\circ} = 18.03 \text{ kN (Compression)}$

$\Sigma F_H = 0 \rightarrow$

$F_{AB} - F_{AC} \cos \theta_1 = 0$

$\therefore F_{AB} = 18.03 \cos 33.69^\circ = 15 \text{ kN (Tension)}$

Joint B: $\Sigma F_V = 0 \rightarrow F_{BC} = 0$

$\Sigma F_H = 0 \rightarrow F_{CD} = F_{BA} = 15 \text{ kN (Compression)}$

Joint C:

$\Sigma \text{Forces normal to } AC = 0 \rightarrow$

$F_{CD} = 0, \text{ since } F_{BC} = 0.$

$\Sigma \text{Forces parallel to } AE \rightarrow$

$F_{CE} = F_{AC} = 18.03 \text{ kN (Compression)}$

Joint D:

$\Sigma F_V = 0 \rightarrow F_{DE} = 0$

$\Sigma F_H = 0 \rightarrow F_{DF} = F_{BD} = 15 \text{ kN (Tension)}$

Joint E:

$\Sigma \text{Forces normal to } CG = 0 \rightarrow$

$F_{EF} = 0$

$\Sigma \text{Forces parallel to } CG \rightarrow$

$F_{EG} = F_{CE} = 18.03 \text{ (Compression)}$

Joint F:

$\Sigma F_V = 0 \rightarrow$

$F_{AG} = 12 \text{ kN (Compression)}$

All member forces are marked on Fig. 4.11 (b).

Example 4.5. Analyse the truss shown in Fig. 4.12 (a). All members are 3 m long.

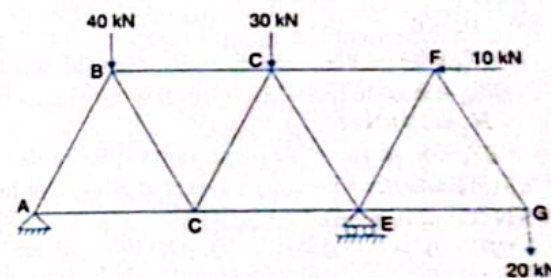


Fig. 4.12 (a)

Solution. Since all members are of same length (3 m), all triangles are equilateral triangles and hence all included angles are 60° . Joint by joint analysis is carried out and the direction of joint forces are marked in Fig. 4.11 (b).